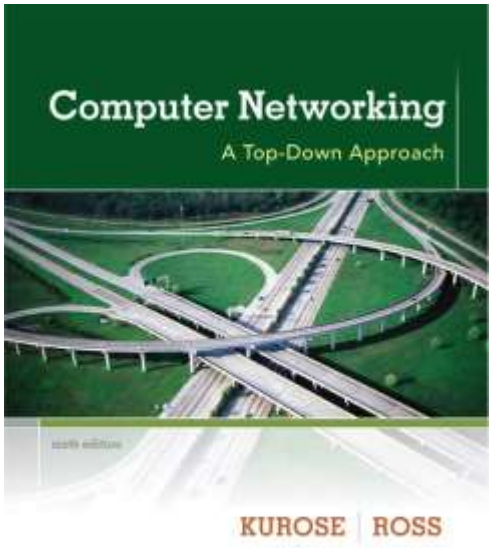
Computer Communications & Networks

CSNC-2413



Week: 12

- Ch 4 : Network Layer

Computer Networking: A Top Down Approach
6th edition
Jim Kurose, Keith Ross
Addison-Wesley
March 2012

Dr Nauman Mazhar

Faculty of Information Technology (FIT)
University of Central Punjab (UCP)

Chapter 4: outline

4.1 introduction

4.2 virtual circuit and datagram networks

4.3 what's inside a router

4.4 IP: Internet Protocol

- datagram format
- IPv4 addressing
- ICMP
- IPv6

4.5 routing algorithms

- link state
- distance vector
- hierarchical routing

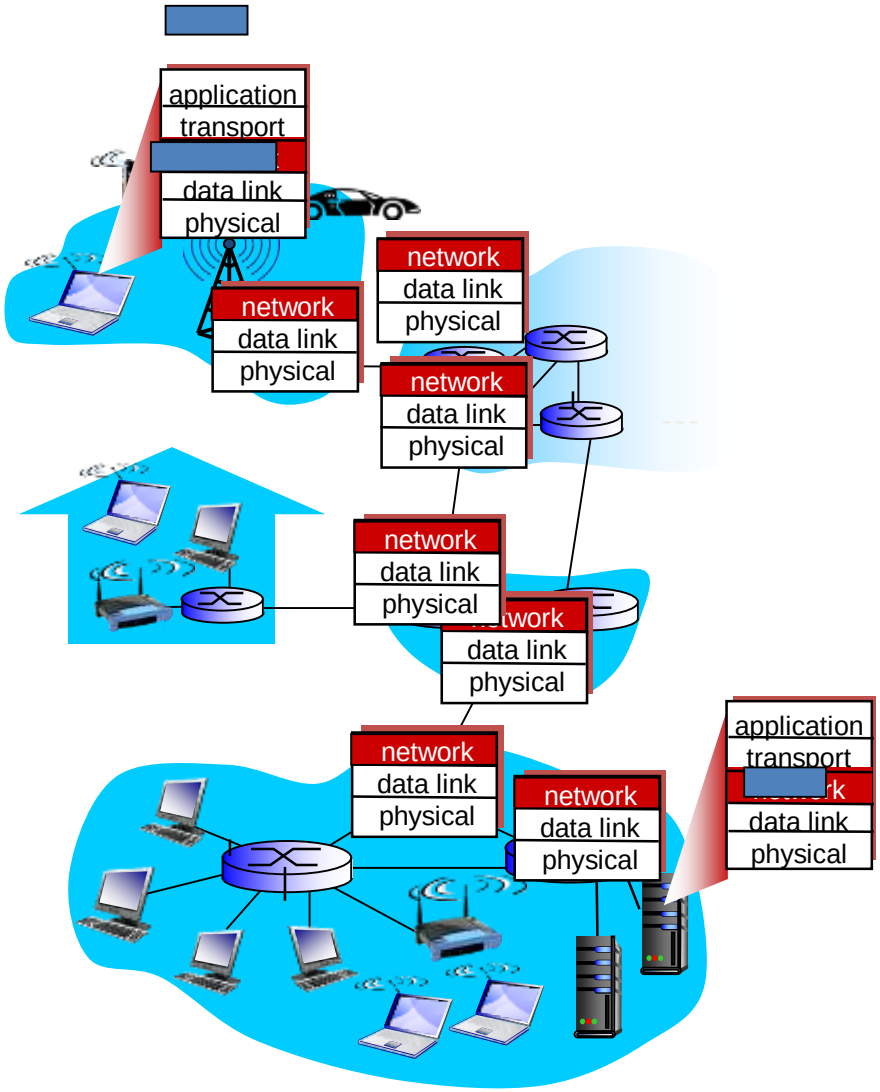
4.6 routing in the Internet

- RIP
- OSPF
- BGP

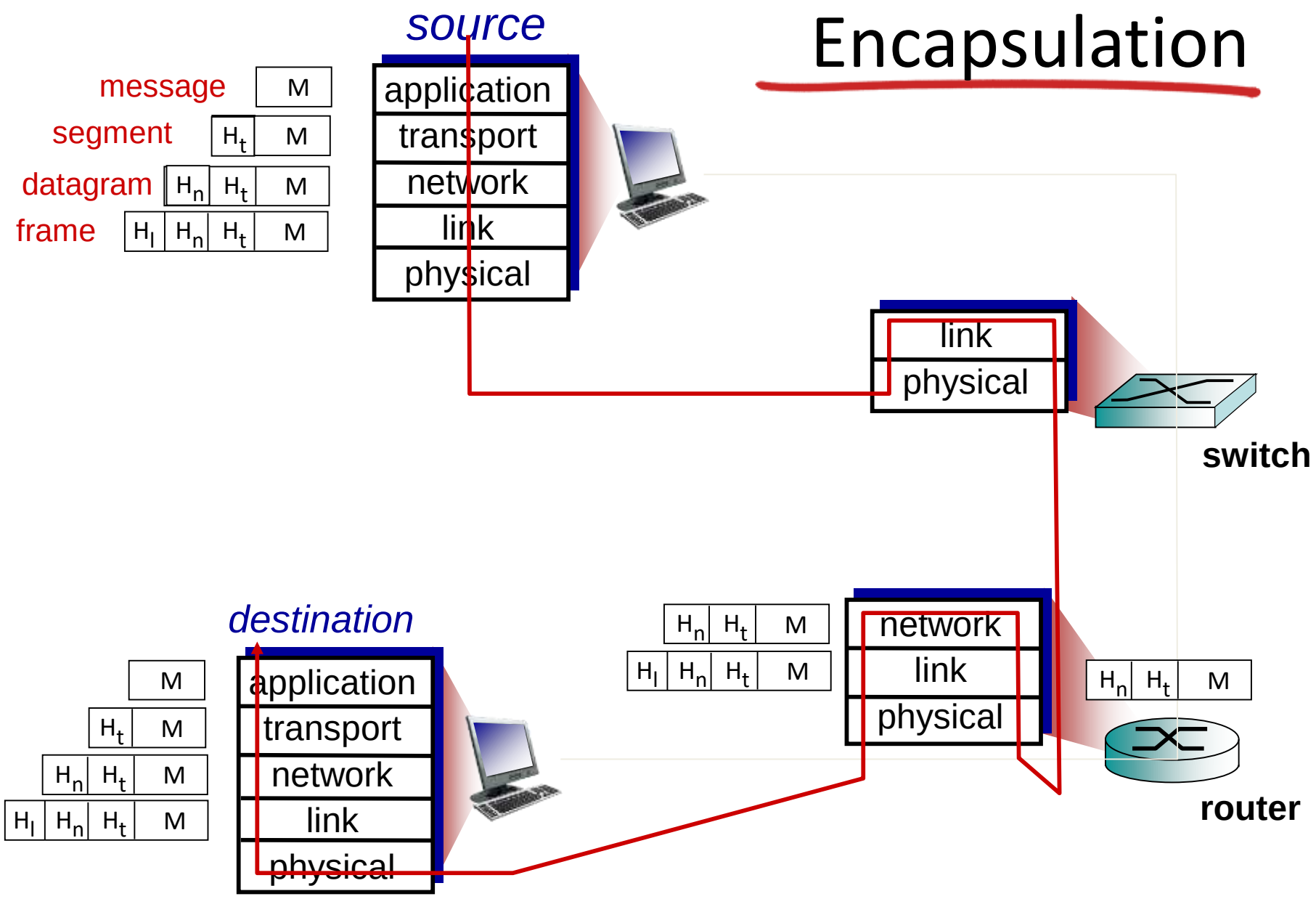
4.7 broadcast and multicast routing

Network layer

- transport IP pkts from sender to receiver ***host***
- sending side:
  encapsulate segments into IP pkts
- receiving side:
  extract segments from IP pkts, deliver to transport layer
- network layer protocols:
  help move IP pkts thru the N/W
- *routers* examine header fields in IP pkts *(upto N/W layer)*

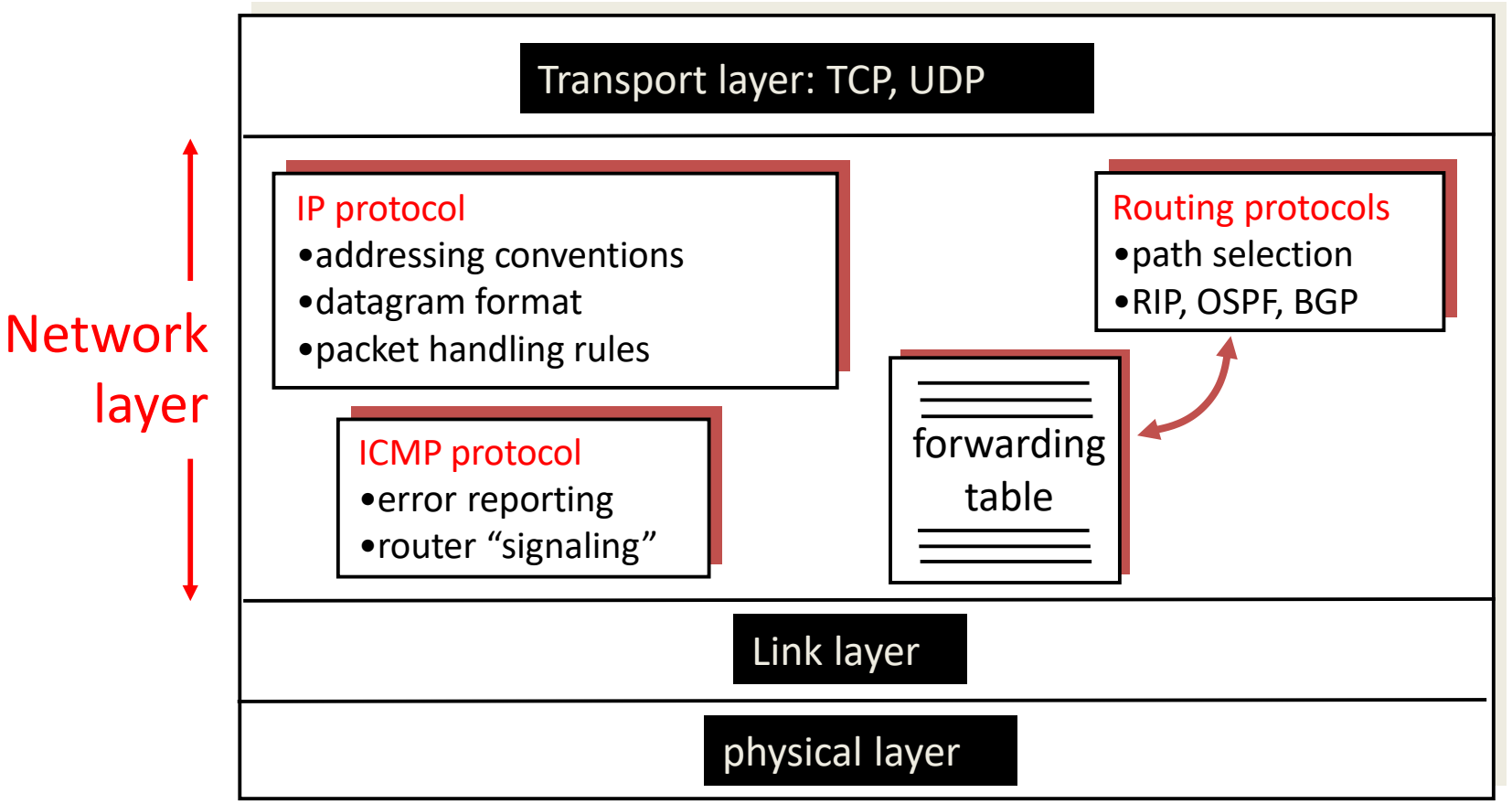


Encapsulation



The Network layer

Network layer functions at hosts/routers :



Two key network-layer functions

- ***Forwarding:***

  move packets from router input port to appropriate output port

- ***Routing:***

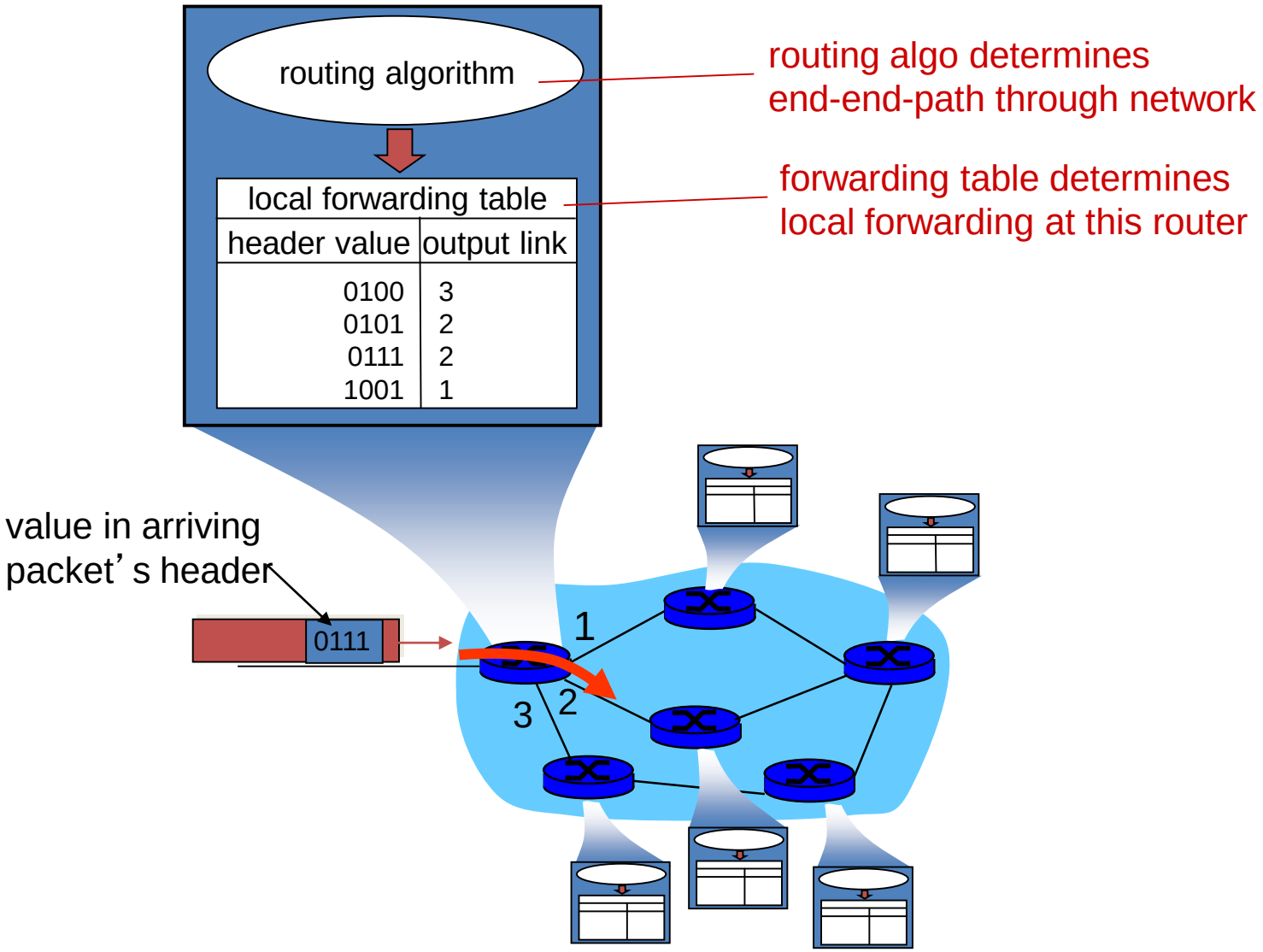
  determine the route taken by packets, from source to destination

  - *run a routing algo*
  - *build routing table*
  - *learn forwarding table*

analogy:

- *routing:* process of planning the trip from source to dest
- *forwarding:* process of getting through a single interchange

Interplay between routing and forwarding



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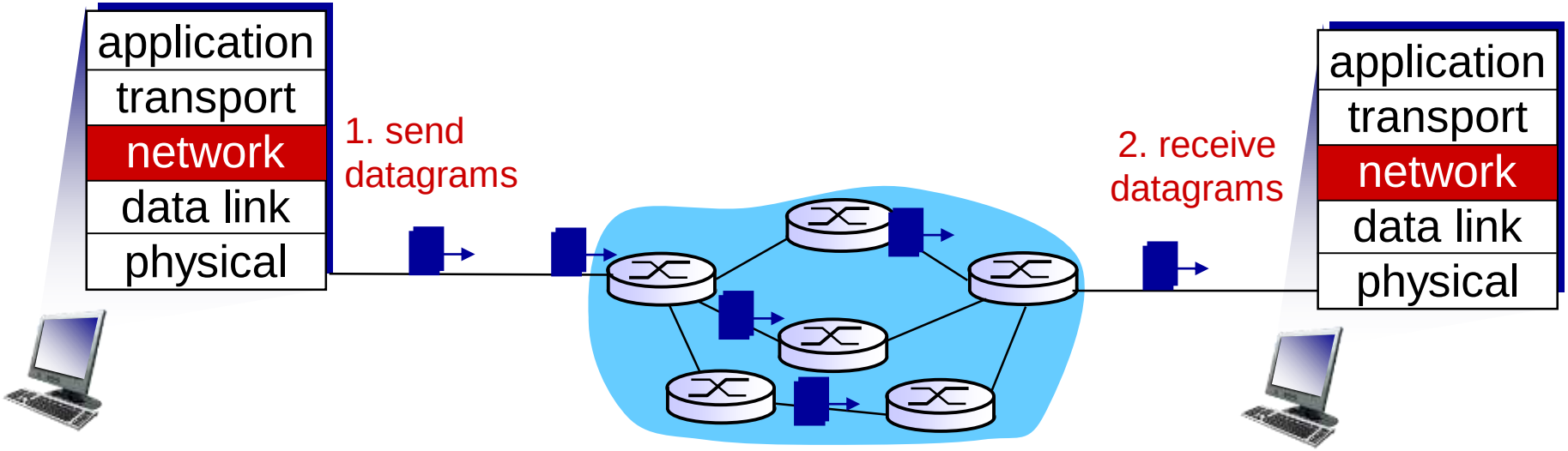
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Network layer service model

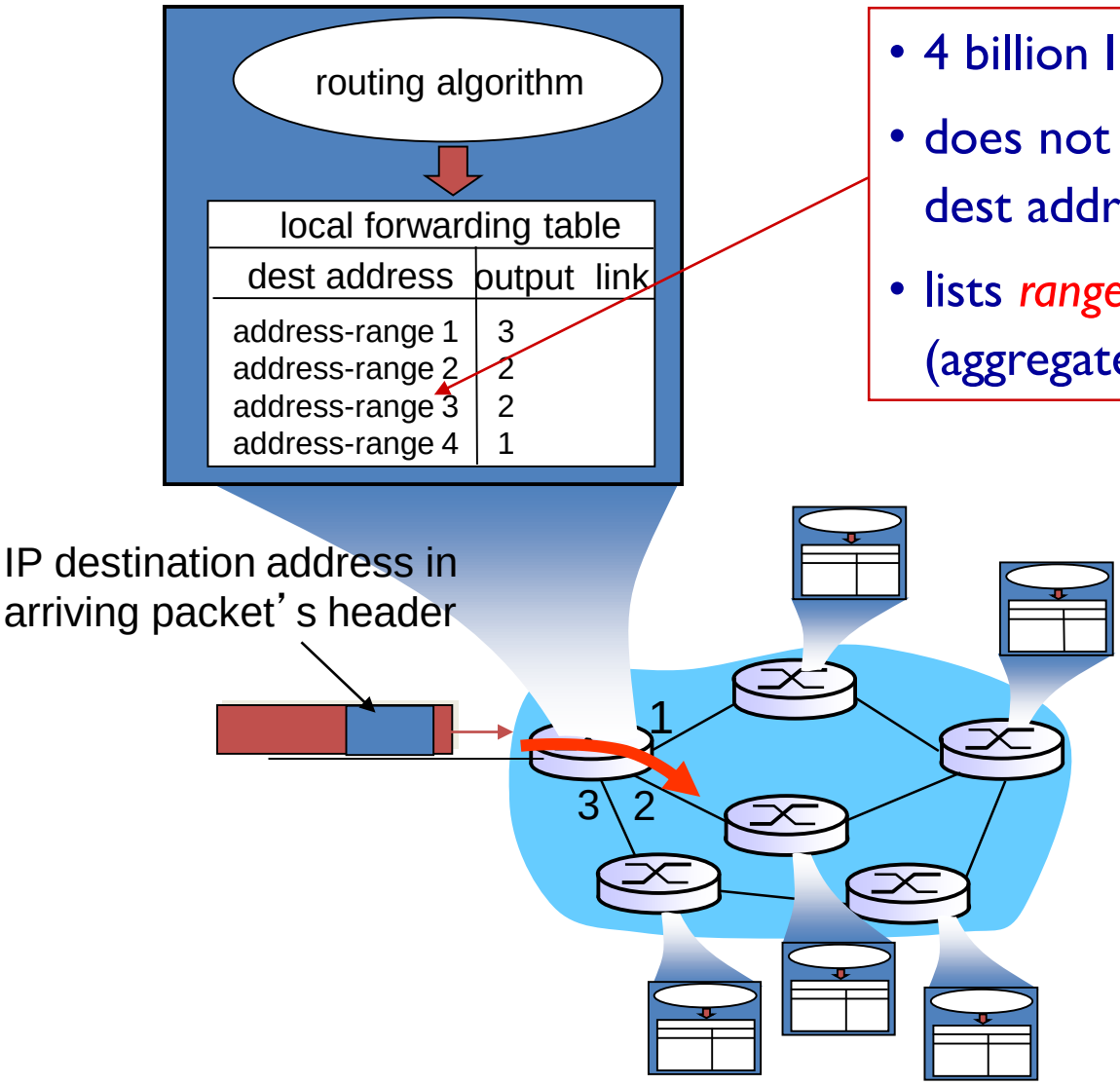
- Connection-oriented or connection-less service
- *Virtual-Circuit Network* :
  provides network-layer *connection oriented* service
  (pkts forwarded using virtual circuit numbers)
- *Datagram Network* :
  provides network-layer *connectionless* service
  (pkts forwarded using dest IP addr)
- Same as TCP/UDP connection-oriented / connectionless service at transport layer, but …
  - *Service* : host to host
  - *no choice* : network provides one or the other
  - *Implementation* : in network core

Datagram networks

- Packets forwarded using address of destination host
- No network-level concept of "connection"
  - No call setup at network layer
  - No state maintained at routers for connections
  - each pkt treated independently



Datagram forwarding table



- 4 billion IP addrs
- does not list individual dest addrs
- lists *range* of IP addrs (aggregate table entries)

Datagram forwarding table

Destination Address Range	Link Interface
11001000 00010111 00010000 00000000 thru 11001000 00010111 00010111 11111111	0
11001000 00010111 00011000 00000000 thru 11001000 00010111 00011000 11111111	1
11001000 00010111 00011001 00000000 thru 11001000 00010111 00011111 11111111	2
otherwise	3

router matches a prefix of pkt dest addr with entries in forwarding table

Longest prefix matching

longest prefix matching

when matching forwarding table entry for given dest addr, use *longest* addr prefix that matches dest addr

Destination Address Range	Link interface
11001000 00010111 00010*** *********	0
11001000 00010111 00011000 *********	1
11001000 00010111 00011*** *********	2
otherwise	3

examples:

DA: 11001000 00010111 00010110 10100001 which interface?

DA: 11001000 00010111 00011000 10101010 which interface?

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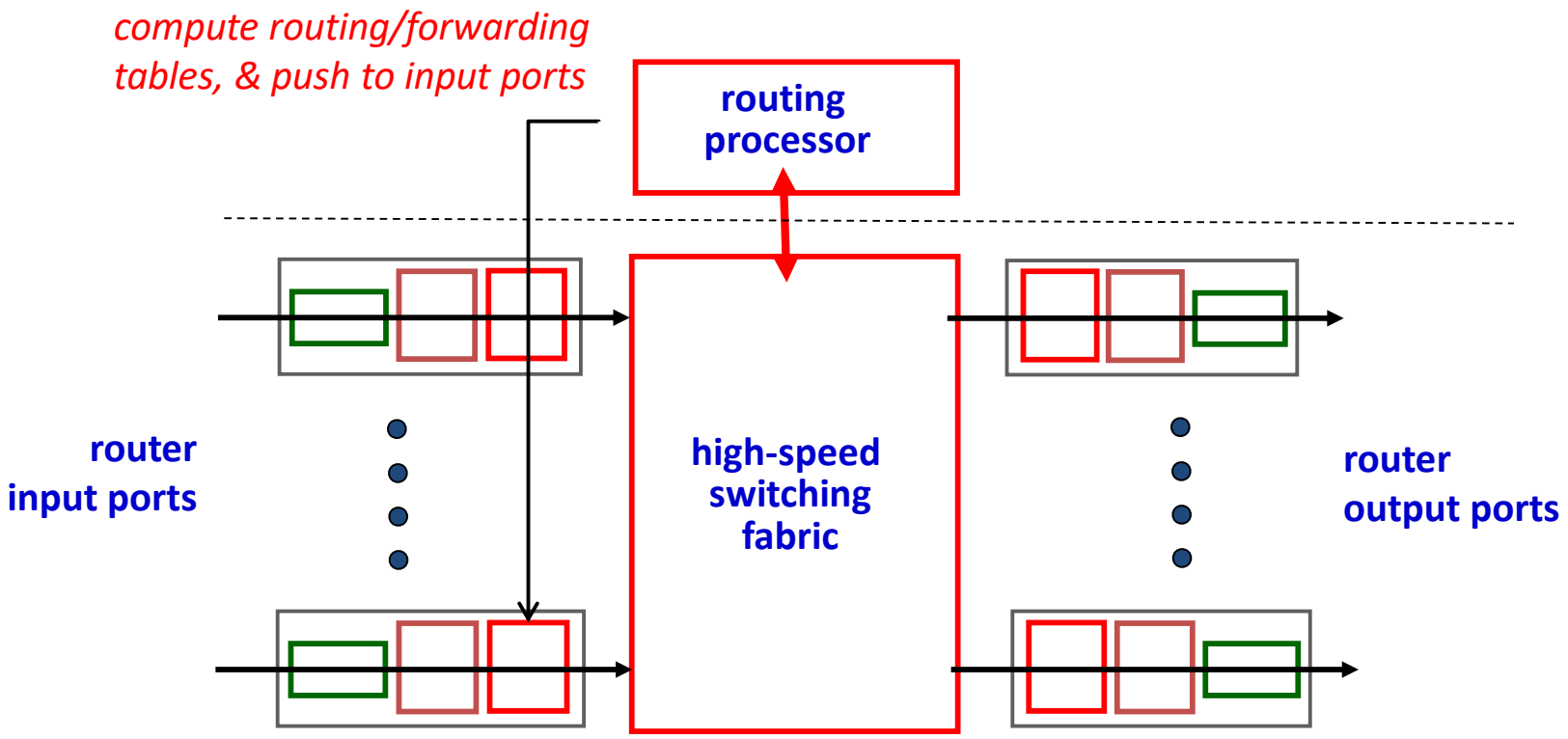
Cisco Router (Front)

Cisco Router (Back)

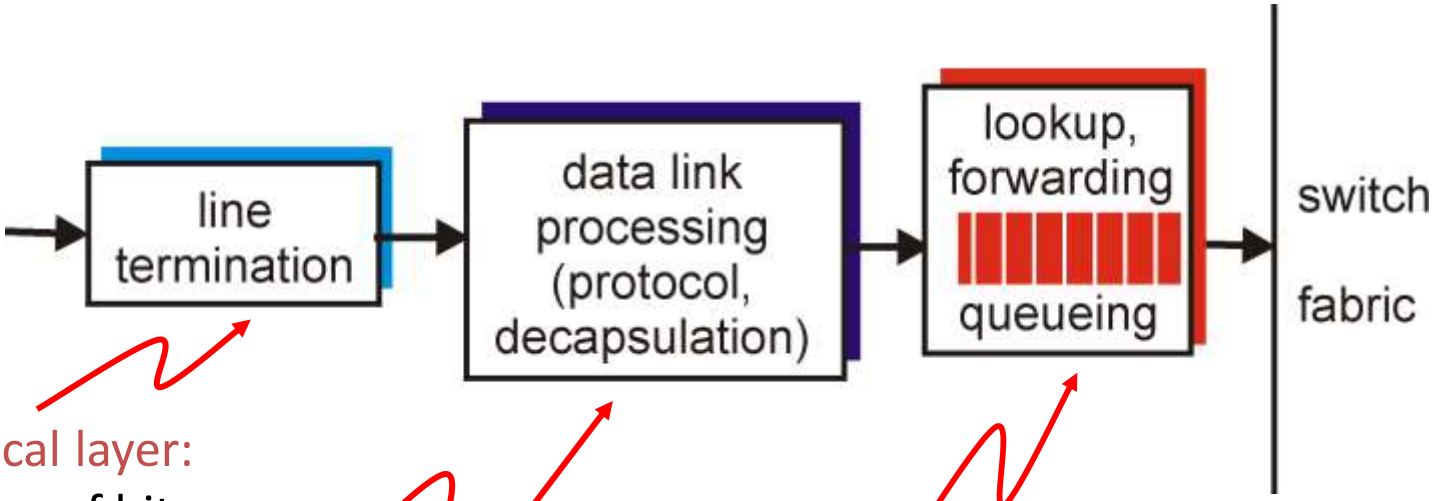
Router architecture overview

Two key router functions:

- run routing algorithms/protocols (RIP, OSPF, BGP)
- process incoming datagrams, & *forward* to outgoing links



Input Port Functions



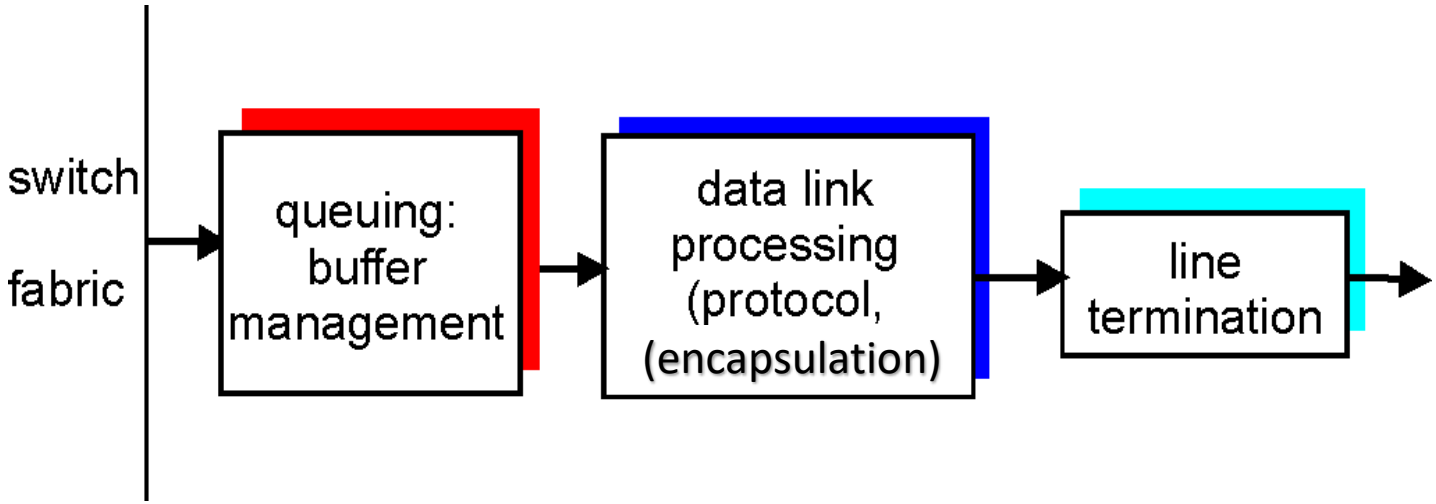
Physical layer:
reception of bits from signals

Data link layer:
e.g, Ethernet protocol functions & header removal

Packet forwarding:

- for given dest addr, lookup output port in forwarding table
- goal: perform lookup/forwarding at 'line speed'
- if arrival rate faster than forwarding into switch fabric - queuing

Output Ports



- *Queuing* required when datagrams arrive from fabric faster than the output data rate
- Protocol encapsulation
- Conversion of bits into signals

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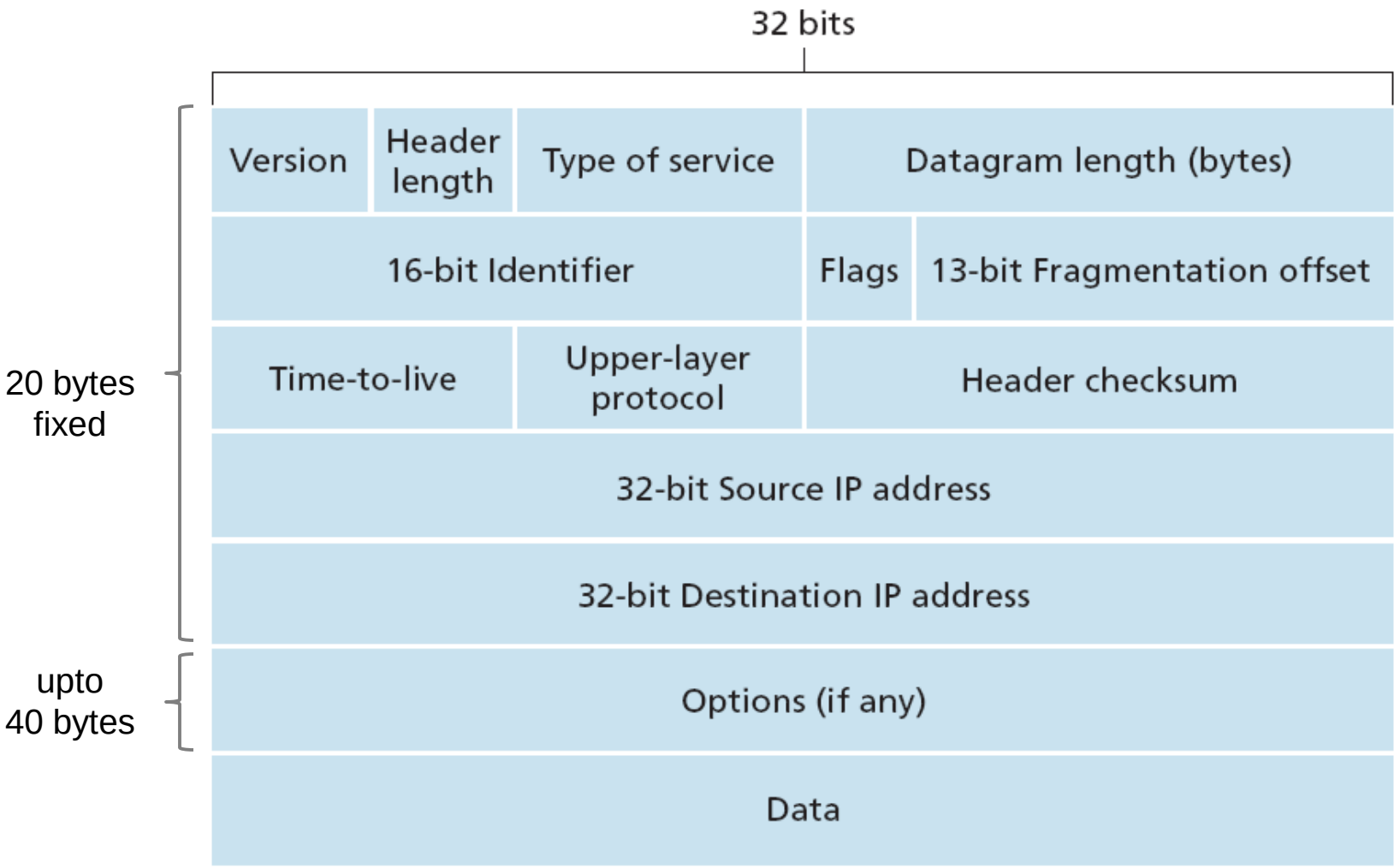
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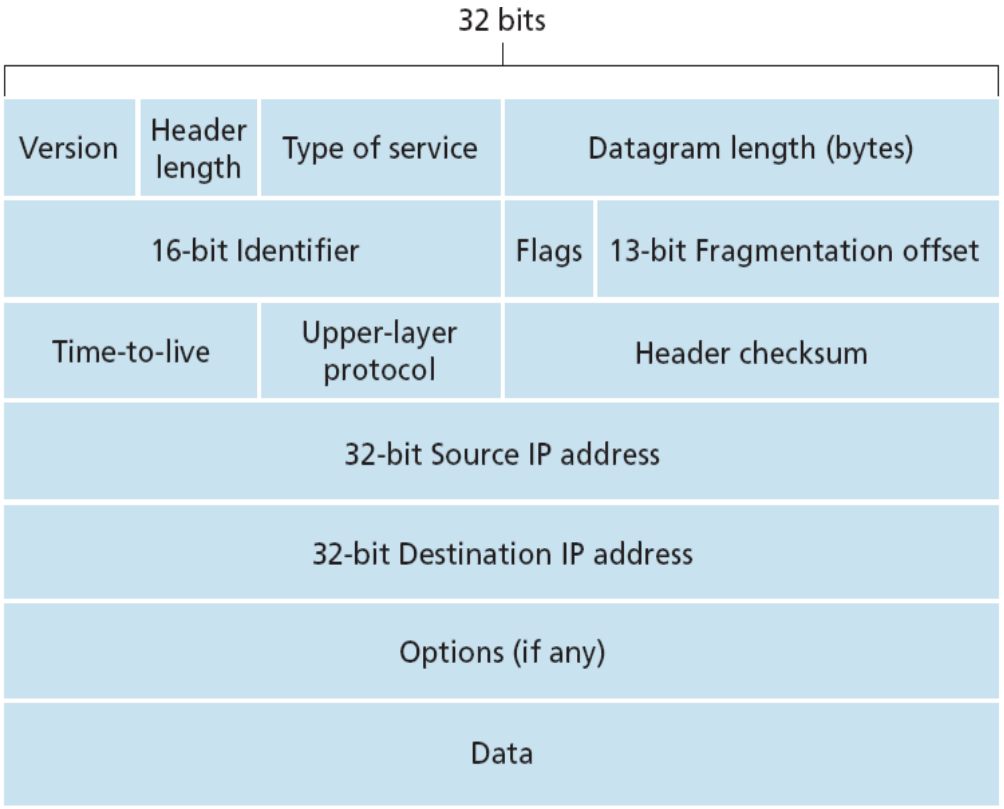
4.7 broadcast and multicast routing

IP datagram format

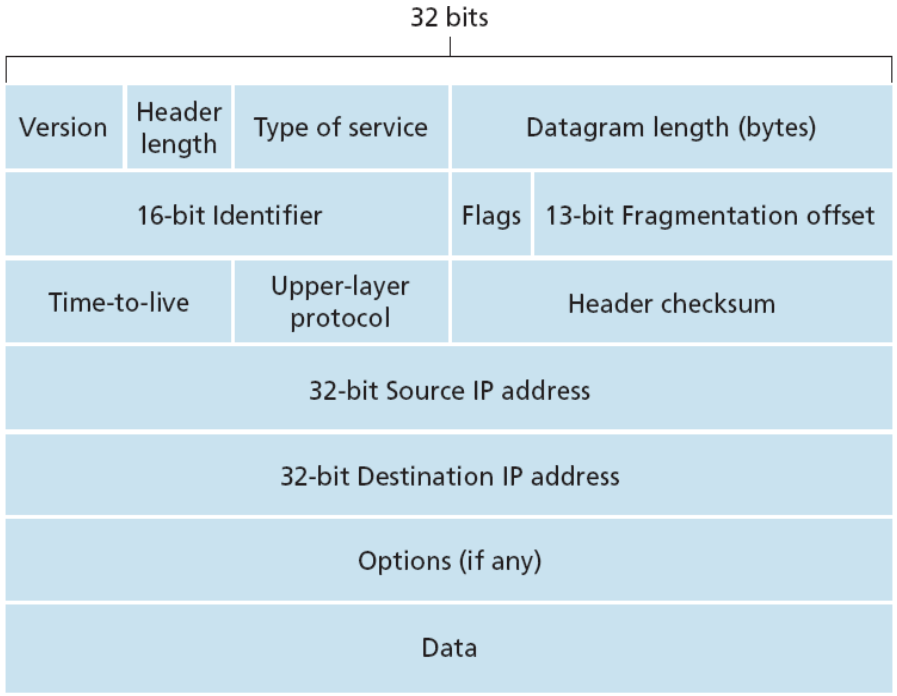


IP header fields

- Version - 4 bit
  - `4` for IPv4
  - `6` for IPv6
  - host/router determines how to interpret datagram
- Header Length - 4 bit
  - in 32 bit words
  - variable (20 – 60 bytes), due to variable number of options
  - min (typical) value is `5` for 20 byte header *(without options)*



IP header fields



- TOS - 8 bit
  - meant to mark pkt priority, for different service types
  - distinguishes IP datagrams *(requiring low delay, high throughput, or reliability)*
- Total Length - 16 bit
  - datagram length in bytes *(header + data)*
  - theoretical max is 65,535 bytes; but rarely larger than 1,500 bytes

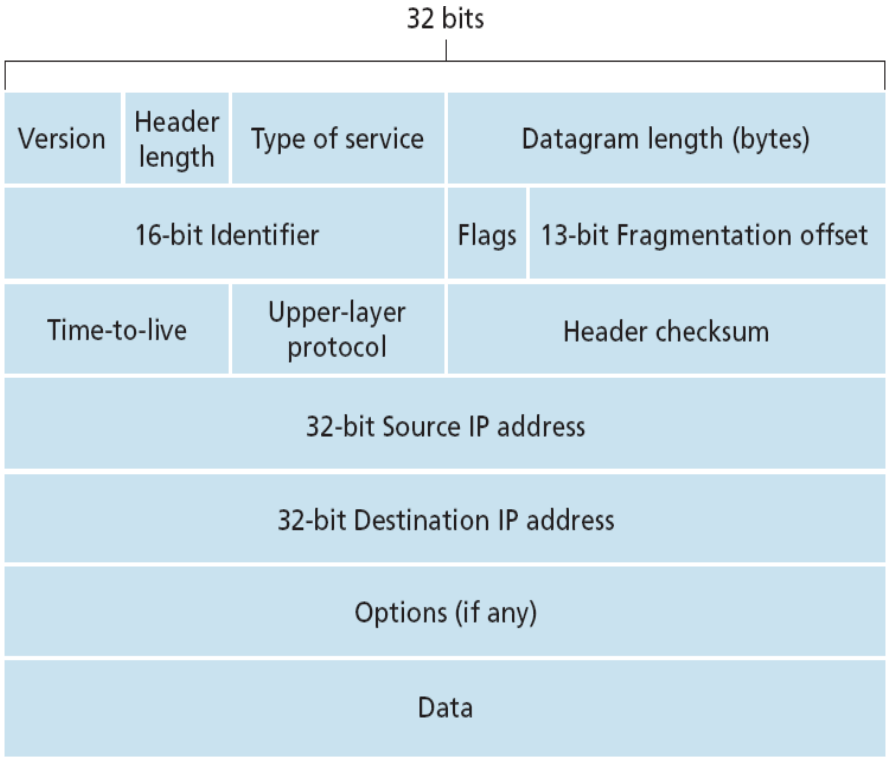
IP header fields

- 16 bit Identification
  - sequence number, used in fragmentation
  - uniquely identifies datagram *(along with src/dest IP addrs, & protocol)*
- 3 bit Flags
  - 1 bit Reserved (*not used*)
  - Don't Fragment bit : prohibits fragmentation
  - More bit : used for fragmentation & re-assembly

    `0' marks last fragment
- 13 bit Fragmentation Offset
  - indicates where in original datagram the fragment belongs
  - measured in 64 bit units (*8 bytes*)
  - implies that fragments other than the last must contain a data field that is multiple of 64 bits

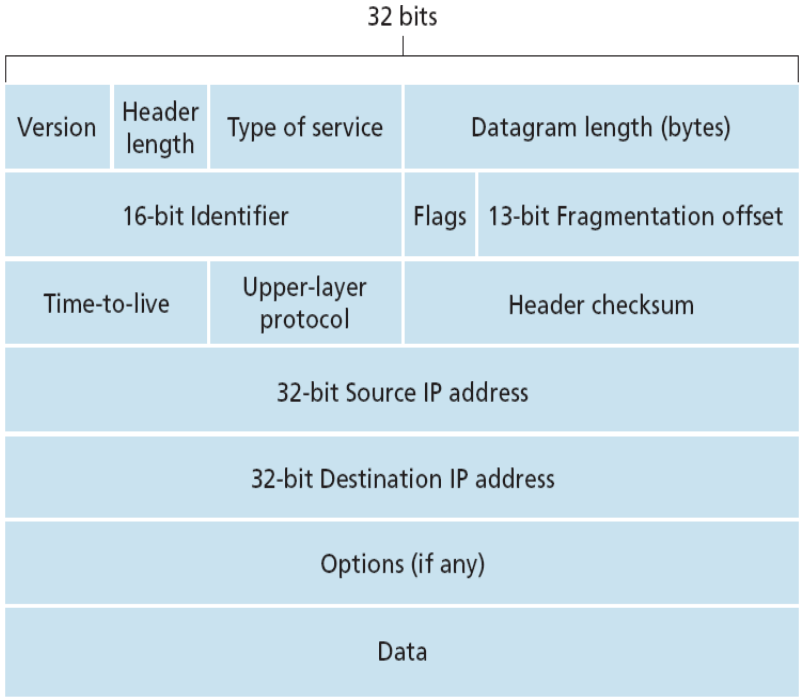
IP header fields

- 8 bit Time to Live
  - max hop count that packet can remain in Internet *(default 64)*
  - each router decrements TTL by 1
  - when TTL = 0, router discards datagram
  - ensures datagrams do not circulate forever *(routing loops)*



- 8 bit Protocol
  - indicates transport layer protocol to receive data; TCP=6, UDP=17
  - used when datagram reaches final dest
  - de-mux key

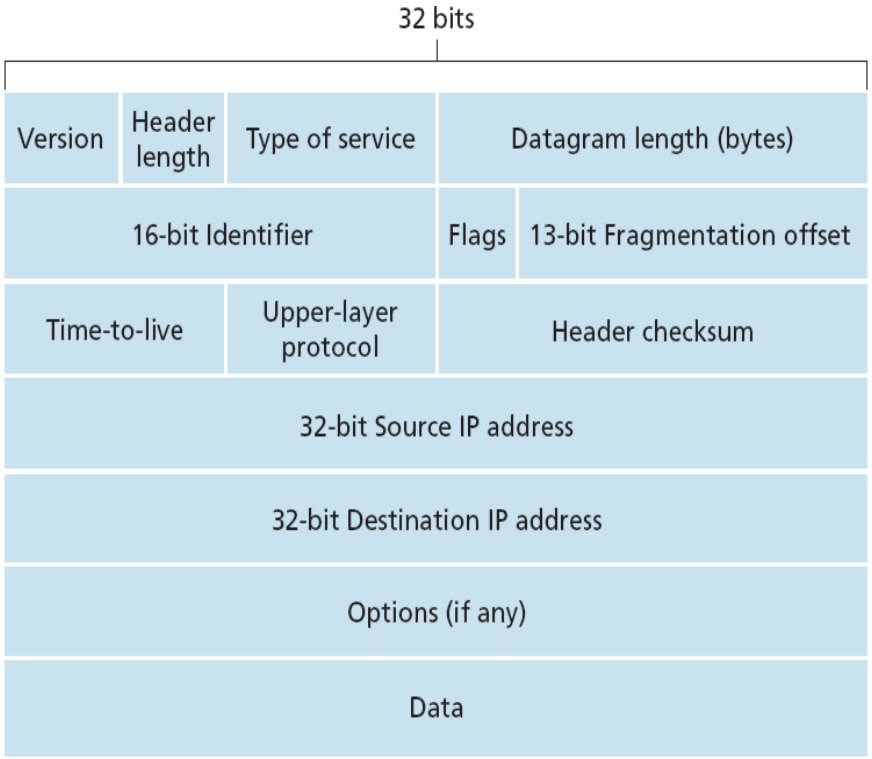
IP header fields



- Header Checksum - 16 bit
  - error detection code applied only to IP hdr

    (detect bit errors in recvd hdr)
  - Checksum computation:
    Treat each 2 bytes in hdr as
    16 bit integers
    Take sum using 1s complement arithmetic
    Then take 1s complement of sum
  - routers verify & re-compute checksum for each received datagram
  - Error condition: if computed checksum does not match the checksum received in hdr, datagram discarded

IP header fields

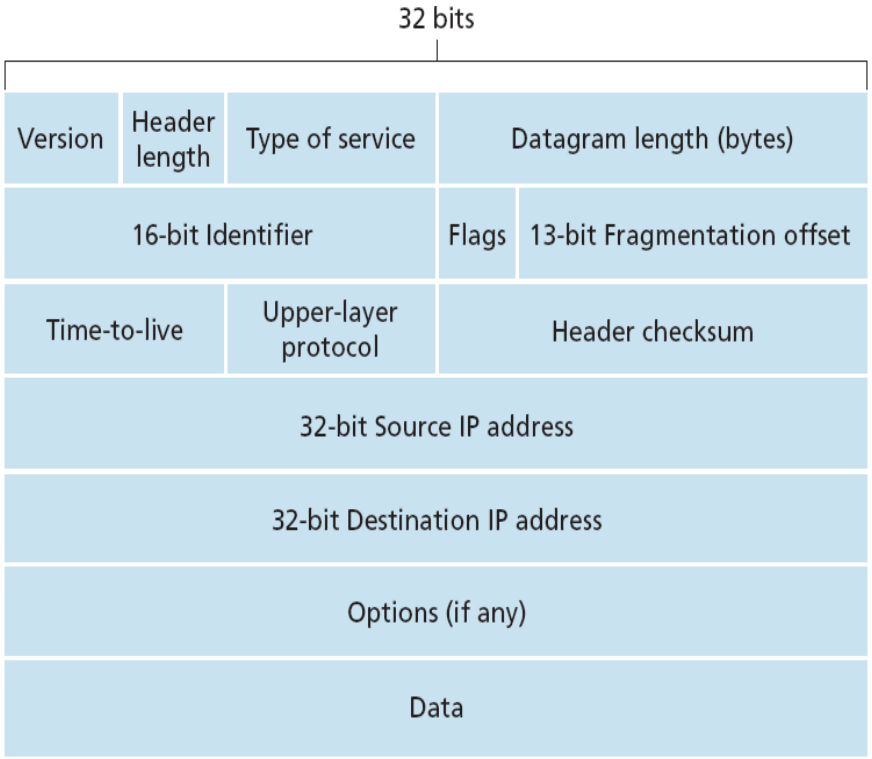


- Source IP & Destination IP Addresses
  - 32 bit addresses
  - when source creates datagram, IP addrs inserted into relevant fields
- Options
  - variable length, max 40 bytes
  - options requested by sender

    e.g, timestamp, record route taken, source routing, etc
  - options used rarely — reduces overhead by not including options fields in every datagram

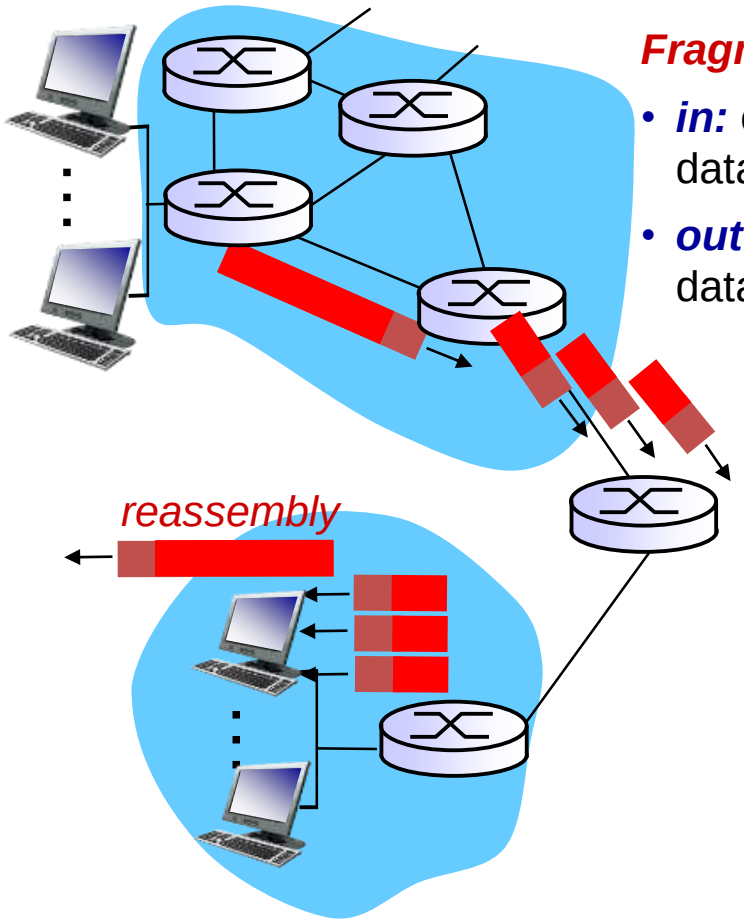
IP header fields

- Padding
  - to fill to multiple of 32 bits long
- Data Field
  - carries transport layer segment (TCP or UDP)
  - can also carry other types of data, such as ICMP messages
  - integer multiple of 8 bits long (bytes)



IP fragmentation, reassembly

- network link has an MTU (max transfer unit) - largest size IP pkt it can carry
  - different links, different MTUs *(Ethernet: 1500 bytes)*
- larger IP datagrams divided ("fragmented") by routers
  - one datagram becomes several datagrams
  - "reassembled" only at final destination
  - IP header bits used to identify & re-order, related fragments



IP fragmentation, reassembly

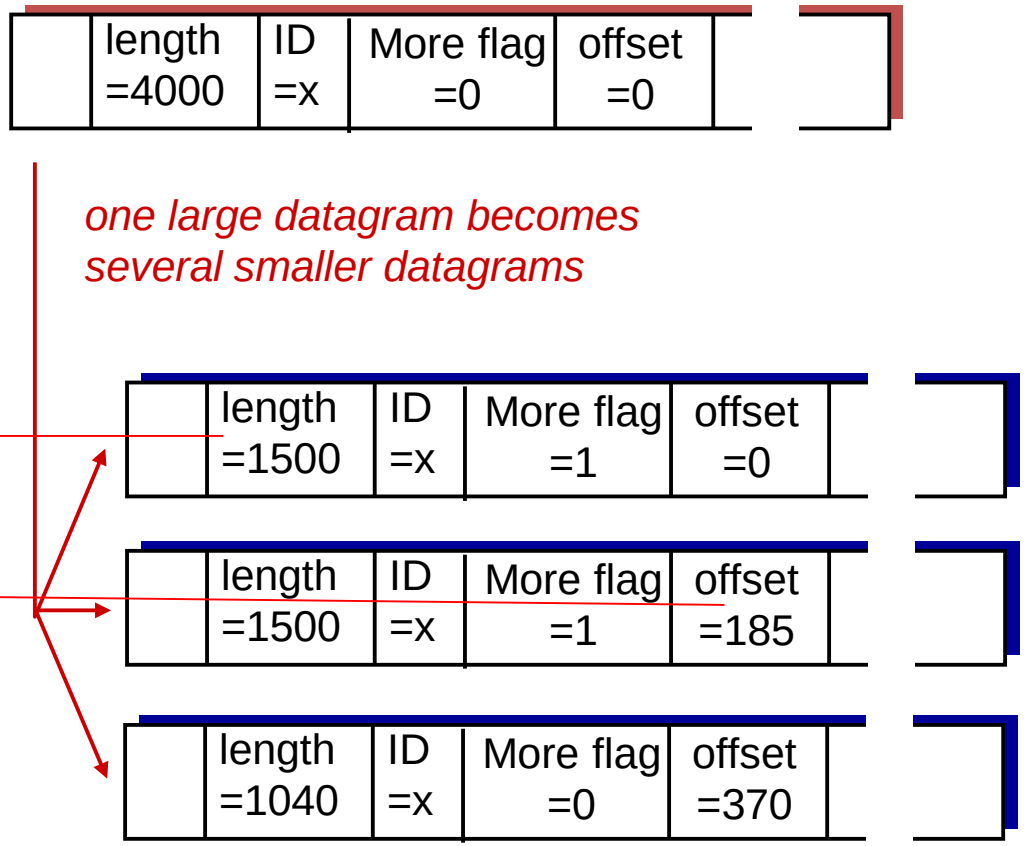
- Datagram of 4,000 bytes (20 bytes of IP header plus 3,980 bytes of IP payload) arrives at a router and must be forwarded to a link with MTU of 1,500 bytes
- Implies that 3,980 IP payload data bytes in original datagram must be allocated to three separate fragments

  (each of which is also an IP datagram)
- Amount of payload data, in all but the last fragment, should be a multiple of 8 bytes, and offset value should be specified in units of 8-byte chunks
- If original datagram is stamped with identification number 777, header field values of 03 fragments are shown in next slide…

IP fragmentation, reassembly

example…

- 4000 byte datagram
- MTU = 1500 bytes
- X = 777



IP fragmentation, reassembly

- When router needs to fragment an IP datagram, each resulting fragment is stamped with source address, destination address, and identification number of original datagram
- When destination receives a series of datagrams from the same sending host, it can examine 'identification' numbers of datagrams to determine which datagrams are actually fragments of the same larger datagram
- Because IP is an unreliable service, one or more fragments may never arrive at destination; all other fragments would need to be dropped

IP fragmentation, reassembly

- In order for destination host to be absolutely sure it has received the last fragment of original datagram, the last fragment has '**More**' flag bit set to **0**, whereas all other fragments have this flag bit set to **1**
- To allow destination host to reassemble fragments in their proper order, the '**Offset**' field is used to specify where the fragment fits within the original IP datagram;
- '**Offset**' field also helps to determine whether some fragment is missing